

TECHARENA

Global Talent Building the Future Together

Italy

Physics-checked AI for RF tolerance analysis

Sub-topic 3 | A 30-minute solver run replaced by a checked surrogate model

One tolerance study costs months of solver time

30 min

one full-wave simulation

per board design

277 days

sensitivity analysis

solver only: 13,312 runs (KPI 2)

110 years

design search

solver only: 1.92 M runs (KPI 1)

0.25-0.40 dB

design margin today

vs 1.3 dB of factory spread

Serial solver time at 30 minutes per run. Tolerance studies need thousands to millions of runs.

Only 3 in 10 boards pass today, and finding out why with the solver alone is not affordable.

A 60-run surrogate stands in for the solver

- Kriging model of the 11 tolerance parameters.
- 60 solver runs are enough: ten random 60-run models reproduce the 540-run results within about 3 points.
- Reciprocity built in, passivity checked on every predicted matrix – confirmed by Huawei as physics-informed.
- One model answers both criteria: sensitivity (KPI 2) and design search (KPI 1).

Six stages, each backed by a measurement

1

Audit the data

reciprocal to $1e-11$; largest singular value 0.995, limit 1

2

Kriging with consistency checks

full-curve model: 0 of 734 M predicted matrices non-passive

3

Validation on unseen boards

93% verdicts right, 90% intervals cover 91%

4

Exact sensitivity analysis

no random sampling, 0.3 s, from 30 real runs

KPI 2

5

Virtual design and tolerance study

961 designs x 2,000 boards = 1.92 M evaluations

KPI 1

6

Scenario check, boards held out

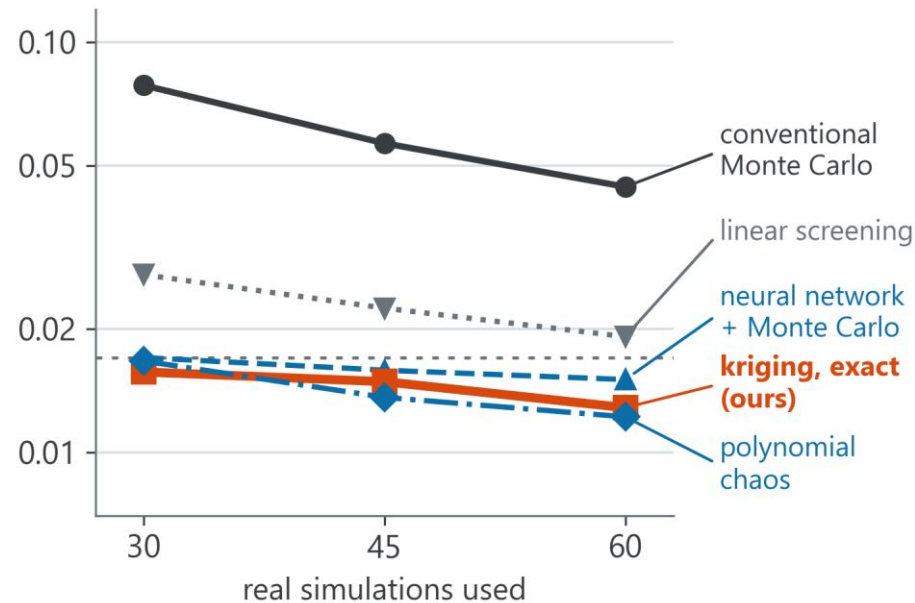
5 of 5 predictions inside the real interval

Across stages 3 to 5: other models, other limits, uncertainty

KPI 2 – sensitivity: the right drivers from 30 runs, not 13,000

Error in the sensitivity indices

lower is better, mean of 3 metrics, 10 random draws
dashed line: noise floor of the 540-run reference



Which errors matter, from 45 real runs

circle and line: estimate with 90% bootstrap interval
diamond: the same analysis on all 540 runs

Reflection

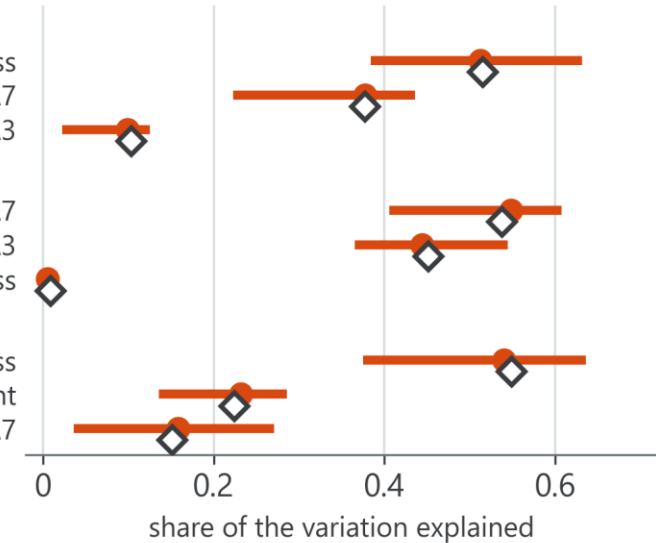
board thickness
copper width A7
copper width A3

Coupling

copper width A7
copper width A3
board thickness

Isolation

board thickness
material constant
copper width A7



three leading drivers per metric, Huawei tolerances, uniform scatter

At equal runs, conventional Monte Carlo is 3 to 5 times less accurate; the three surrogates tie.

KPI 1 – design search: conventional versus this pipeline

Conventional: solver in the loop

Which errors matter



277 days

How many boards pass



12 days

Design search, 1.92 M evaluations



110 years

13,312 + 561 + 1.92 million solver runs, log scale

This pipeline: one 60-run surrogate

Which errors matter



15 hours

How many boards pass



20 hours

Design search, 1.92 M evaluations



30 h + 5 min

one model from 60 simulations answers all three, same log scale

Serial times at 30 minutes per run. With 100 solver licences in parallel, the design search alone still takes about a year.

Validation: from 30% to 85% of boards passing

Each scenario's real boards were held out, the model retrained, and its prediction compared with them: 5 of 5 inside the real interval.

Last step: 85% predicted, 73% observed on 30 boards – consistent, not proof.

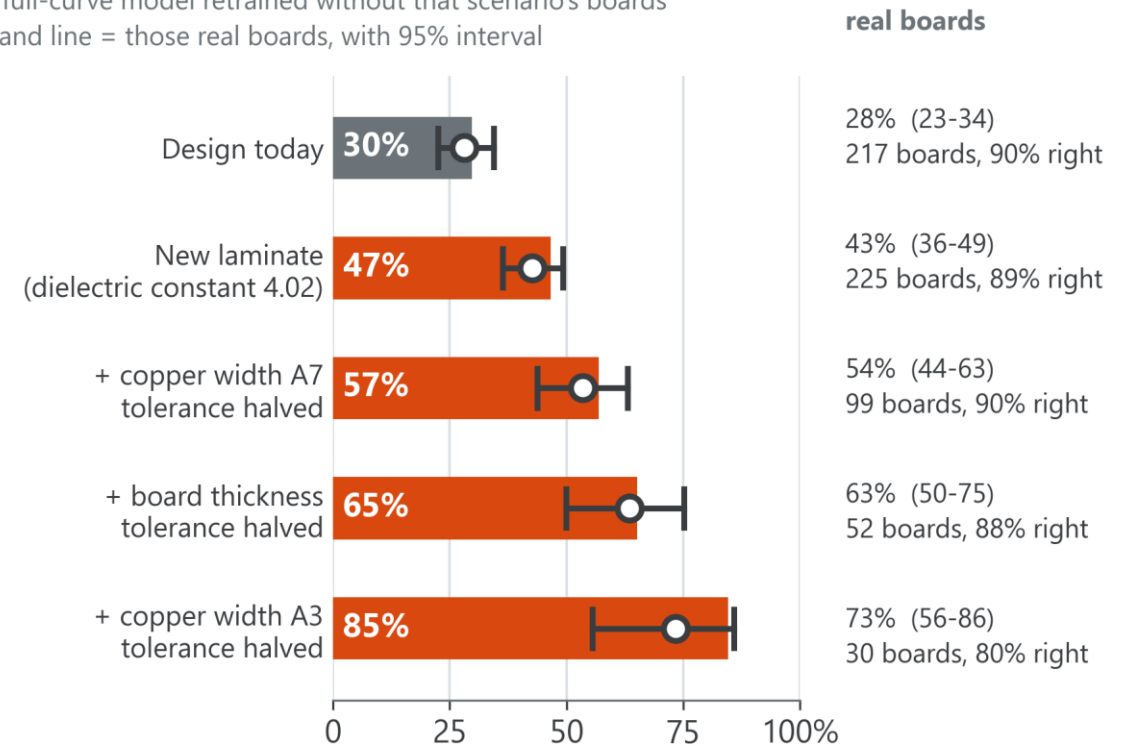
Pass/fail agreement stays 92-97% when the illustrative limits are halved or tripled.

Open: model bias, the confidential stack-up, no extra solver runs.

0 of 734 million predicted matrices break passivity.

Boards passing under the stated limits

bar = full-curve model retrained without that scenario's boards
circle and line = those real boards, with 95% interval



each step keeps the ones above it; every predicted matrix checked for passivity

last step: 30 real boards; the 11-point gap is within sampling noise

State of the art, and where this differs

	Common practice in the literature	This project
Surrogate	Kriging / Gaussian process, about a third of RF surrogate studies [4]	Kriging, checked against polynomial chaos and a neural network on the same boards
Output	Compress the frequency response first (PCA or response features)	No compression: PCA raised the error 2.8x on this data, so all 3,012 outputs are predicted
Physics	Physics-informed networks put Maxwell's equations in the loss; they need field data [6]	Only port data exists: reciprocity built in, passivity checked on 734 million matrices
Sensitivity	Closed-form Sobol indices from a GP [1-3]; kriging-PCE in EM dosimetry [5]	Applied to the 11 manufacturing tolerances; top drivers right from 30 runs
Yield	GP-assisted Monte Carlo that re-checks borderline boards in the solver [7]	No extra runs allowed: every pass rate checked against held-out real boards

[1] Oakley & O'Hagan (2004), J. R. Stat. Soc. B 66(3)

[2] Marrel et al. (2009), Reliab. Eng. Syst. Saf. 94(3)

[3] Le Gratiet, Cannamela & Iooss (2014), SIAM/ASA J. Uncertain. Quantif.

[4] Systematic review, surrogate-based EM design of RF components (2026), Sensors 26(8):2504

[5] Kersaudy et al. (2015), J. Comput. Phys. 286:103-117

[6] Khan & Lowther (2022), IEEE Trans. Magn. 58(9)

[7] Fuhrländer & Schöps (2020), J. Math. Ind. 10

Project folder

Scan to open the full material

README: pipeline, evidence and limitations

Code that reproduces every number

Figures and the architecture report

QR code goes here

run `insert_qr.py` with your folder link



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THANK YOU

From 277 days of solver time to 15 hours – with the physics checked on every prediction.

Next step we ask for: the real acceptance limits and a handful of verification runs at the proposed design.